Backbone-correct answers retained (%)
Mathematical training
Reasoning Gym
Sequential training
SFT 11.9 22.6 5.2 6.6 41.6 42.1 54.1 89.7 48.6 77.2 72.4 71.8 28.4 66.1 65.5 71.2
LoRA 6.4 19.2 8.1 16.4 40.4 50.8 55.2 78.0 31.7 58.4 62.5 55.0 25.2 63.7 67.7 68.9
BT 21.6 81.0 15.5 32.0 66.6 73.8 28.1 9.8 20.2 85.3 29.4 64.1 20.6 87.3 36.4 70.0
BGT 27.5 74.5 8.5 22.0 20.5 40.6 33.1 46.1 11.5 82.2 42.8 78.3 14.2 84.4 47.7 85.7
Longshot 72.0 37.3 56.3 60.8 74.6 70.2 69.9 80.4 79.4 90.1 62.0 74.7 61.0 82.9 76.2 82.8
0 20 40 60 80 100
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 2B on
Qwen 4B off
Qwen 4B on
Qwen 27B off
Qwen 27B on
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 4B off
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 4B off
Backbone-wrong answers recovered (%)
SFT 1.4 1.4 1.0 1.3 3.5 9.3 7.0 21.6 5.2 9.2 10.9 8.2 3.9 4.2 8.3 7.8
LoRA 0.7 1.9 2.7 4.7 4.5 10.0 6.7 16.2 4.4 6.6 7.2 4.3 2.9 4.3 8.3 9.0
BT 3.8 6.5 2.2 4.7 3.3 8.0 1.6 2.5 2.8 5.5 2.8 5.7 3.0 5.8 6.6 6.9
BGT 5.1 5.4 2.1 3.1 1.2 5.8 1.6 8.0 2.0 4.9 4.2 7.3 2.2 6.4 4.3 7.1
Longshot 5.2 6.2 6.3 8.9 6.1 10.9 4.9 15.5 2.6 5.2 4.8 4.7 4.5 7.3 8.3 7.1
0 5 10 15 20 25
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 2B on
Qwen 4B off
Qwen 4B on
Qwen 27B off
Qwen 27B on
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 4B off
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 4B off